8.033 Fall 2025: Problem Set 8 Solutions

1.

(a) Let's verify this directly by computing the value for g

$$\begin{aligned} 1g &= \frac{GM_{earth}}{R_{earth}^2} = \frac{6.67 \cdot 10^{-11} m^3 kg^{-1} s^{-2} \times 5.97 \cdot 10^{24} kg}{(6.37 \cdot 10^6 m)^2} = 9.81 ms^{-2} \\ &= 9.81 \frac{m}{s^2} \frac{1ly}{3 \cdot 10^8 m/s \times 3.15 \cdot 10^7 s} \times \left(\frac{3.15 \cdot 10^7 s}{1yr} \right)^2 \\ &= 1.03 ly/yr^2 \approx 1 ly/yr^2. \end{aligned} \tag{26}$$

(b) We have seen how this is done in the lecture, where we derived the proper acceleration of a friend being a constant distance L_0 away from the top of a rocket. Using the same formula to calculate the proper acceleration at the bottom of the rocket, we have

$$\alpha_b = \frac{\alpha_t}{1 - L_0 \alpha_t / c^2}, \tag{27}$$

where I have inserted factors of c to make the dimension correct. Plugging in $\alpha_t = 1ly/yr^2$, $L_0 = 1/2ly$, and $c = 1ly/yr$, we find $\alpha_b = 2ly/yr^2$.

(c) Suppose the top of the rocket was located at $x = 1/\alpha_t$ and the bottom of the rocket was located at $x = 1/\alpha_t - L_0$ at the instant ($t = 0$) when the rocket started accelerating from rest. The spacetime diagram is shown in Fig. (3).

(d) We have seen in class that the top and bottom of the rocket share the same line of simultaneity, so the measurement of proper time happens over the same range of boost parameter, namely we have $\alpha_t \tau_t = \alpha_b \tau_b$. To see this explicitly, consider trajectories of the top and bottom of the rocket, as seen in Fig. (1)

$$t_t(\tau_t) = \frac{1}{\alpha_t} \sinh(\alpha_t \tau_t), \;\; x_t(\tau_t) = \frac{1}{\alpha_t} \cosh(\alpha_t \tau_t) \tag{28}$$

$$t_b(\tau_b) = \frac{1}{\alpha_b} \sinh(\alpha_b \tau_b), \;\; x_b(\tau_b) = \frac{1}{\alpha_b} \cosh(\alpha_b \tau_b). \tag{29}$$

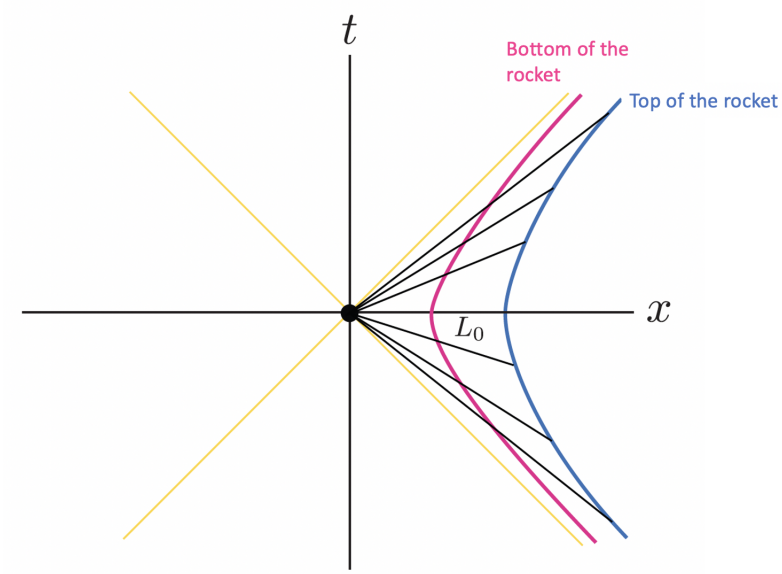

Figure 1: Spacetime diagram of the rocket seen in the MCRF at the beginning of the acceleration. The yellow solid lines are light cones from the origin (not required to show) and the solid black lines that pass through the worldlines of the top and bottom of the rocket

Suppose a line of simultaneity goes through the top of the rocket's trajectory at $(t_t(\tau_t'), x_t(\tau_t'))$ and intersects the bottom of the rocket's trajectory at $(t_b(\tau_b'), x_b(\tau_b'))$. This means that the clock at the top of the rocket reads $\tau_t = \tau_t'$, while at the same instant the clock at the bottom of the rocket reads $\tau_b = \tau_b'$. Since $(t_t(\tau_t'), x_t(\tau_t'))$ and $(t_b(\tau_b'), x_b(\tau_b'))$ lie on the same line that goes through the origin, we have

$$\frac{t_t(\tau_t')}{x_t(\tau_t')} = \frac{t_b(\tau_b')}{x_b(\tau_b')}. \tag{30}$$

Using Eq. (28) and (29), we get

$$\tanh(\alpha_t \tau_t') = \tanh(\alpha_b \tau_b'), \tag{31}$$

so

$$\alpha_t \tau_t' = \alpha_b \tau_b', \tag{32}$$

which means

$$\tau_t' = \frac{\alpha_b}{\alpha_t}\tau_b' = 2\tau_b'. \tag{33}$$

In other words, every second that passes for a clock at the bottom of the rocket, two seconds passes for you!

(e) We use Eq. (27) again

$$\alpha_b' = \frac{\alpha_t}{1 - L_0'\alpha_t/c^2}. \tag{34}$$

Using $L_0' = (99/100)ly$ this time and $\alpha_t = 1ly/yr^2$, we get $\alpha_b' = 100ly/yr^2$.

2.

(a) In class, we have seen that a merry-go-round of radius R in an inertial frame still has radius R in the frame that is spinning with the merry-go-round. However, the circumference measured in the spinning frame will be $2\pi R\gamma$. Applying this to the problem, we conclude that a builder at a distance D from the hub measures the circumference to be

$$C(D) = 2\pi D\gamma(v(D)), \tag{1}$$

where $v(D)$ is the tangential velocity, which can be found from the angular velocity by

$$v(D) = \omega D = \frac{0.8c}{R}D. \tag{2}$$

So, Eq. (1) becomes

$$C(D) = 2\pi D\frac{1}{\sqrt{1-(0.8c/R)^2D^2}}. \tag{3}$$

(b) Consider a builder located at point p on the platform (see Fig. [1]). We can find a MCRF of p, and the builder's proper time $d\tau$ is equal to the infinitesimal time element dt_{MCRF} at that instant. We know how time dilation works between the MCRF and the ground inertial reference frame

$$dt = \gamma(v)dt_{MCRF}, \tag{4}$$

where dt is the infinitesimal time element in the ground inertial reference frame. Then we know at the instant of p, the builder's proper time and the time in the ground frame are related by

$$dt = \gamma(v)d\tau. \tag{5}$$

Due to symmetry around the hub (as long as the builder maintains the distance D from the hub), Eq. (5) is always true around the hub. Therefore, we can relate the time passed in the ground frame with time measured by a builder located at a distance D from the hub as

$$\tau(D) = (\sqrt{1-(0.8c/R)^2D^2})t. \tag{6}$$

In other words, after one day has passed in the ground frame, only $\sqrt{1-(0.8c/R)^2D^2}$ of a day has passed for the builder. The further away the builder from the hub, the younger they will be at the end of the day in the ground frame.

(c) When builders are constructing the platform at a distance D away from the hub, the total proper length of all materials needed to pave around the hub is just the circumference, which is $2\pi D\gamma$ as seen in part (a). At the instant when the platform stops, the materials that were originally at a distance D away from the hub are still at a distance D away from the hub, but the circumference now reduces to $2\pi D$. At the same time, the proper length of the materials does not change. Since $2\pi D\gamma > 2\pi D$, there is too little room for all the material at a distance D from the hub. Therefore, the platform will have to expand outward for the material to fit.

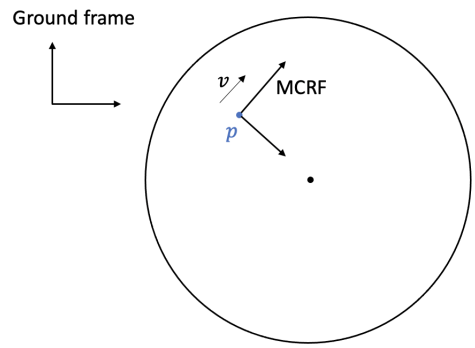

Figure 1: Suppose the builder is at point p. Then we can find a MCRF that moves with speed $v = \omega D$ (see Eq. [2]) relative to the ground reference frame at that instant.

3.

(a) We have already found the line element in spherical coordinates in pset 4 problem 2. Here we only need three dimensions, so the line element is

$$ds^2 = -dt^2 + dr^2 + r^2 d\theta^2. \tag{7}$$

(b) Suppose the merry-go-round has a radius r and the observer started at an angle θ at $t = 0$, their coordinates in the ground inertial reference frame parametrized by the time t is

$$\begin{aligned} t &= t, \\ r(t) &= r, \\ \theta(t) &= \theta - \omega t. \end{aligned} \tag{8}$$

(c) With the new definition of the coordinates, Eq. (8) becomes

$$\begin{aligned} t' &= t, \\ r' &= r, \\ \theta' &= \theta - \omega t. \end{aligned} \tag{9}$$

Under this new coordinates, the line element Eq. (7) is

$$\begin{aligned} ds^2 &= -(dt')^2 + (dr')^2 + (r')^2(d\theta' + \omega dt')^2 \\ &= -[1 - (r')^2\omega^2](dt')^2 + (dr')^2 + (r')^2(d\theta')^2 + 2(r')^2\omega d\theta' dt'. \end{aligned} \tag{10}$$

(If you need a review on how we derived the line element in the primed coordinates, take a look at pset 4 problem 1 part (c).)

Since

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu, \tag{11}$$

we can compare Eq. (11) with (10) to find the metric components in this coordinate system. We find the non-zero components are

$$g_{00} = -[1 - (r')^2\omega^2],\ g_{11} = 1,\ g_{22} = (r')^2,\ g_{02} = g_{20} = (r')^2\omega, \tag{12}$$

where index numbers 0, 1, and 2 represent t', r', and θ', respectively. All the remaining components are zero. Note that we have ensured the symmetry of the metric by enforcing $g_{02} = g_{20} = (r')^2\omega$.

(d) Intuitively, since we have defined $t' = t$, it is actually the same as the time measured by an inertial observer in the ground frame. Since the merry-go-round frame is rotating relative to the ground frame, the intuition of relativity that we have built so far tells us that an observer rotating along with the merry-go-round should experience time differently than an inertial observer in the ground frame, so events that are simultaneous in the ground frame are not necessarily simultaneous in the merry-go-round frame.

To see this more precisely, notice that we found non-zero off-diagonal terms in the metric, namely $g_{02} = g_{20} = (r')^2\omega$. This term mixes space and time, meaning that whenever we advance in θ' along the circle, we also advance in t'. So to measure a single simultaneous length, we have to synchronize all the infinitesimal lengths along the circle.

(e) Alice computes half the trip time by taking the total time she waits for the light ray to arrive back and dividing it by 2. This suggests the halfway point for Alice is $((dt'_1+dt'_2)/2, \theta'_A)$, where we have omitted the r' coordinate since all observers we consider are at the edge of the merry-go-round ($r' = r$). Bob computes the halfway point by measuring the time at which he receives the light ray. This suggests the halfway point for Bob is $(dt'_1, \theta'_A + d\theta')$. Alice and Bob's halfway points are simultaneous. This means two infinitesimally close clocks on the disk are synchronized if the coordinate time difference between them is

$$dt' = dt'_1 - \frac{1}{2}(dt'_1 + dt'_2) = \frac{1}{2}(dt'_1 - dt'_2). \tag{13}$$

(f) First, consider the null path from $(0, \theta'_A)$ to $(dt'_1, \theta'_A + d\theta')$, namely the path of light emited by Alice and received by Bob. Using $ds^2 = 0$ and the second line of Eq. (10), with $r' = r$, we get

$$-[1 - r^2\omega^2](dt'_1)^2 + r^2(d\theta')^2 + 2r^2\omega d\theta' dt'_1 = 0. \tag{14}$$

Solve this quadratic equation of dt'_1, we get

$$dt'_1 = \frac{rd\theta'(\omega r \pm 1)}{1 - r^2\omega^2}. \tag{15}$$

Recall that ωr is the tangential velocity which is less than 1 ($c = 1$), so we discard the solution with the minus sign since it would indicate a negative time interval, so we get

$$dt'_1 = \frac{rd\theta'(\omega r + 1)}{1 - r^2\omega^2}. \tag{16}$$

Now, let's consider the light path from Bob to Alice, that is from $(dt'_1, \theta'_A + d\theta')$ to $(dt'_1 + dt'_2, \theta'_A)$. The null path once again gives us a quadratic equation now for dt'_2

$$-[1 - r^2\omega^2](dt'_2)^2 + r^2(d\theta')^2 - 2r^2\omega d\theta' dt'_2 = 0. \tag{17}$$

We solve for dt'_2 and get

$$dt'_2 = \frac{rd\theta'(1 - \omega r)}{1 - r^2\omega^2}, \tag{18}$$

which is the positive solution.

Plug Eq. (16) and (18) into Eq. (13), we get

$$dt' = \frac{\omega r^2}{1-\omega^2 r^2} d\theta'. \tag{19}$$

This means whenever we move forward $d\theta'$ along the circle, we have to fast forward $\frac{\omega r^2}{1-\omega^2 r^2} d\theta'$ in the coordinate time as well to be synchronized with the previous point. Now we integrate along θ' to find the circumference of the merry-go-round

$$\begin{aligned}
C &= \int \sqrt{ds^2} = \int \{-[1-r^2\omega^2](dt')^2 + r^2(d\theta')^2 + 2r^2\omega dt' d\theta'\}^{1/2} \\
&= \int \left\{ -[1-r^2\omega^2] \left(\frac{\omega r^2}{1-\omega^2 r^2} d\theta' \right)^2 + r^2(d\theta')^2 + 2r^2\omega \left(\frac{\omega r^2}{1-\omega^2 r^2} d\theta' \right) d\theta' \right\}^{1/2} \\
&= \int \left[\frac{\omega^2 r^4}{1-\omega^2 r^2} (d\theta')^2 + r^2 (d\theta')^2 \right]^{1/2} \\
&= \int \left[\frac{r^2}{1-\omega^2 r^2} (d\theta')^2 \right]^{1/2} \\
&= \frac{r}{\sqrt{1-\omega^2 r^2}} \int_0^{2\pi} d\theta' = 2\pi r \gamma,
\end{aligned} \tag{20}$$

where $\gamma = 1/\sqrt{1-\omega^2 r^2}$. We have reached the same conclusion as the one we have already seen in class.